Database Systems Exam Prep Cheat Sheet

Definitions

- **Database:** An organized collection of related data (Lecture 1).
- **DBMS (Database Management System):** Software that manages databases, providing data storage, retrieval, and integrity functions (Lecture 1).
- **Schema / Instance:** *Schema* = the logical design of a database (changed rarely); *Instance* = the data stored at a particular moment (Lecture 2).
- **Data Model:** A collection of concepts for describing data, relationships, and constraints (Lecture 1).
- **Entity:** An object or thing in the real world that can be uniquely identified (Lecture 3).
- **Attribute:** A property or characteristic of an entity. Can be *simple (atomic)*, *composite* (made of subparts), *multi-valued*, or *derived* (Lecture 3).
- **Relationship:** An association among two or more entities. Relationships have cardinality constraints (1:1, 1:N, M:N) and participation (total vs. partial) (Lecture 3–4).
- **Primary Key:** The chosen candidate key that uniquely identifies each tuple (record) in a relation (Lecture 2). For a weak entity, the primary key is the combination of its *partial key* and the primary key of its owner (Lecture 3).
- **Candidate Key:** A minimal superkey; a superkey such that no proper subset is also a superkey (Lecture 2).
- **Superkey:** An attribute or set of attributes that uniquely identifies a tuple in a relation (Lecture 2).
- **Foreign Key:** An attribute (or set) in one relation that references the primary key of another (or the same) relation, creating a link between tables (Lecture 2).
- **Partial Key (Discriminator):** In a weak entity, an attribute (or set) that uniquely identifies weak entities related to the same owner entity (Lecture 3).
- **Weak Entity:** An entity that cannot be identified by its own attributes alone; it must reference a strong (owner) entity. It has a partial key (Lecture 3).
- **Specialization / Subclass:** A hierarchy relationship where a super-entity type is divided into one or more sub-entity types (Lecture 4).

- **Generalization / Superclass:** The inverse of specialization; common features of several subtypes are generalized into a higher-level entity (Lecture 4).
- **Category (Union Type):** A grouping of entity subsets (possibly overlapping) into a single superclass (Lecture 4).

Concepts

- **Levels of DB Architecture:** *Internal* (physical storage), *Logical* (schema), and *External* (user views) schema levels (Lectures 1–2).
- **Data Independence:** The capacity to change schema at one level without altering the schema at the next higher level (Lecture 1).
- **Normalization:** Process of organizing data to minimize redundancy. Definitions:
- **1NF:** All attributes have atomic (indivisible) values; no repeating groups (Lecture 10).
- **2NF:** In 1NF and every non-key attribute is fully functionally dependent on the entire primary key (no partial dependencies) (Lecture 10).
- **3NF:** In 2NF and no non-key attribute is transitively dependent on the primary key (Lecture 10).
- **BCNF:** For every functional dependency $X \rightarrow Y$, X must be a superkey (Lecture 9).
- **Functional Dependency (FD):** A constraint $X \rightarrow Y$ means that if two tuples agree on attributes X, they must agree on attributes Y (Lecture 9).
- **Integrity Constraints:** Rules to ensure data validity. Includes **Entity Integrity** (no primary key is NULL) and **Referential Integrity** (foreign keys must match primary keys in related tables) (Lecture 2–4).
- **ER Diagram Concepts:** Entities (rectangles), attributes (ovals), relationships (diamonds), and cardinalities (Lecture 3–4). Weak entities are shown with double-border symbols (Lecture 3).
- **SQL Data Models: Relational Model**: data stored in tables (relations) of rows and columns (Lectures 2).
- **Join Operations:** Combine tables on matching columns. Types: *INNER JOIN*, *LEFT/RIGHT OUTER JOIN*, *FULL OUTER JOIN*, *CROSS JOIN*, *NATURAL JOIN* (Lecture 7).
- **Set Operations:** *UNION*, *INTERSECT*, *MINUS* (set difference) combine query results (Lecture 7).

SQL Cheat Sheet

SELECT Query: SELECT <columns> FROM <table1> [JOIN <table2> ON <condition>] WHERE <condition> GROUP BY <columns> HAVING <condition> ORDER BY <columns>;

- *Example (Lecture 5/6/11):*
- SELECT Fname, Lname, Address FROM EMPLOYEE, DEPARTMENT WHERE EMPLOYEE.Dno = DEPARTMENT.Dnumber; (Example of basic SELECT with join).
- SELECT SupplierName, City, Country FROM Suppliers; (Lecture 6).
- SELECT * FROM EMPLOYEE INNER JOIN DEPARTMENT ON EMPLOYEE.Dno = DEPARTMENT.Dnumber; (Lecture 7 style join).
- **INSERT:** Add new rows. Syntax: INSERT INTO <table> (<cols>) VALUES (<values>);
- **UPDATE:** Modify rows. Syntax: UPDATE <table> SET <col>=<value>, ... WHERE <condition>; (Lectures 5–7).
- **DELETE:** Remove rows. Syntax: DELETE FROM <table> WHERE <condition>; (Lectures 5–7).
- **CREATE TABLE:** Define a new table. Example:

```
CREATE TABLE EMPLOYEE (
  SSN CHAR(9) PRIMARY KEY,
  Fname VARCHAR(15),
  Lname VARCHAR(15),
  Dnumber INT,
  FOREIGN KEY (Dnumber) REFERENCES DEPARTMENT(Dnumber)
);
```

- **DROP / ALTER:**
- DROP TABLE <table>; – Deletes a table (Lecture 5).
- ALTER TABLE <table> ADD/DROP <constraint>; – Modify table schema.
- **JOIN Variants:**
- INNER JOIN: rows with matching keys in both tables.
- LEFT/RIGHT OUTER JOIN: include all rows from left/right table and matched rows from other (fill unmatched with NULL).
- FULL OUTER JOIN: all rows from both tables.
- **GROUP BY / HAVING / Aggregates:**
- Aggregates: SUM(), COUNT(), AVG(), MAX(), MIN() (Lecture 6).
- GROUP BY: group rows for aggregation.
- HAVING: filter groups (Lecture 6).

- *Example:* `SELECT Dept, AVG(Salary) FROM Employee GROUP BY Dept HAVING AVG(Salary) > 50000;`
- **Distinct:** Eliminate duplicates. `SELECT DISTINCT column FROM table;` (Lecture 8).
- **Subqueries:** Nested queries. Use in WHERE or FROM.
- `ANY/ALL/EXISTS` with subqueries (Lecture 8). *Example:*

```
SELECT Fname FROM EMPLOYEE
WHERE Salary > ANY (SELECT Salary FROM DEPARTMENT WHERE DeptName
= 'Sales');
```

- **SET Operators:**
- `UNION`: combine results of two queries (duplicates removed) (Lecture 7).
- `INTERSECT`: common results (if supported).
- `MINUS` or `EXCEPT`: difference of results (Lecture 7).

Keys and Constraints

- **Primary Key:** Uniquely identifies tuples; implicitly `UNIQUE NOT NULL` (Lecture 2).
- **Foreign Key:** Enforces referential integrity between tables (Lecture 2).
- **Candidate Key:** Any minimal unique identifier.
- **Super Key:** Any superset of a key that is still unique.
- **Composite Key:** A key consisting of multiple attributes.
- **NOT NULL:** Attribute must have a value (Lecture 5–6).
- **UNIQUE:** Values must be distinct (Lecture 2–3).
- **CHECK:** Condition must hold for all tuples (Lecture 10). *Example:* `CHECK (Age >= 0)` to prevent invalid data.
- **Default, Index, etc.:** If mentioned in slides (Indexing in later lectures if any).

ER Diagrams and Modeling

- **Entities and Attributes:** Shown with Entity rectangle and Attribute ovals (Lecture 3).
- **Relationship:** Diamond between entities, labeled with role names and cardinalities (Lecture 3).
- **Weak Entity:** Double rectangle; identified by owning strong entity via an identifying relationship (double diamond) (Lecture 3).
- **Specialization/Generalization:** Triangle with sub-entities under a super-entity (Lecture 4).

- **Constraints:** Total vs. partial participation (bar vs. circle), cardinality (1 or N).

Normalization Concepts

- **1NF:** Eliminate multi-valued/repeating fields (atomicity) (Lecture 10).
- **2NF:** Remove partial dependencies (no attribute depends on part of a composite key) (Lecture 10).
- **3NF:** Remove transitive dependencies (non-key attributes depend only on key) (Lecture 10).
- **Boyce-Codd NF (BCNF):** Stronger than 3NF; every determinant must be a key (Lecture 9 mentions BCNF).
- **Functional Dependency (X → Y):** If X determines Y, then each X-value is associated with exactly one Y-value (Lecture 9).
- **Partial Dependency:** A non-key attribute depends on part of a composite key (Lecture 10).
- **Transitive Dependency:** A non-key attribute depends on another non-key attribute (via the key) (Lecture 10).
- **Decomposition:** Splitting a relation into smaller ones to achieve normal forms (implied by normalization lecture).

Important Comparison Tables

- **DDL vs DML vs DCL:**
- DDL (Data Definition Language): `CREATE`, `ALTER`, `DROP`, etc. (definition of structure) (Lecture 5).
- DML (Data Manipulation Language): `SELECT`, `INSERT`, `UPDATE`, `DELETE` (data queries/updates) (Lecture 5).
- DCL (Data Control Language): `GRANT`, `REVOKE` (access privileges) (Lecture 5).
- **Entity vs. Weak Entity:** (Lecture 3)

Aspect	Entity	Weak Entity
Identification	Own primary key	Cannot be identified by own attributes; uses partial key + owner PK
Key	Primary key	Partial key + owner key
Relationship	May exist alone	Existence depends on owning entity

- **Superkey vs Candidate vs Primary:** (Lecture 2)

Term	Definition
Superkey	Any set of attributes uniquely identifying tuples (may be non-minimal)
Candidate Key	Minimal superkey (no redundant attributes)
Primary Key	Selected candidate key for the relation

Exam Tricks (Common Pitfalls)

- **SQL:** Remember that `NULL` is not equal to anything; use `IS NULL` or `IS NOT NULL` (Lecture 5–7).
- **JOIN vs WHERE:** Inner joins can be written using `JOIN ... ON` or listing tables with conditions in `WHERE`. Be careful with outer joins (Lecture 7).
- **Normalization:** Always verify the definition: e.g., 2NF only applies if there is a composite key (Lecture 10).
- **Keys:** A table with a single-attribute key is automatically in 2NF (Lecture 10 note).
- **Case Sensitivity:** SQL keywords are case-insensitive, but string comparisons may be case-sensitive depending on the DB (implied from SQL practice).
- **ANY/ALL:** `ANY` returns true if any value meets the condition, `ALL` requires all values meet it (Lecture 8). Misusing them often leads to wrong results.

Past Exams Analysis

Frequently Tested Topics

- **SQL Queries:** Many exams focus on writing SELECT statements (basic queries, joins, subqueries, aggregates).
- **Normalization:** Questions on normal forms (1NF, 2NF, 3NF) and functional dependencies appear in multiple exams.
- **ER Modeling:** Converting ER diagrams to relational schemas or identifying entity/relationship types.
- **Keys and Constraints:** Identifying primary/foreign keys, enforcing integrity.

- **Joins and Set Operations:** Usage of different JOIN types and UNION/INTERSECT in queries.
- **Aggregate Queries:** GROUP BY/HAVING usage is common (Lecture 6 patterns).

Instructor’s Favorite Topics

- **Composite Keys and Partial Dependencies:** Emphasized in normalization questions (Lecture 10).
- **Subqueries (ANY/ALL):** Rasha’s lectures cover these, and similar patterns likely tested (Lecture 8).
- **Weak Entities:** Dr. Ibrahim highlighted weak entities and composite primary keys (Lecture 3).
- **SQL Constraint Syntax:** Questions on defining tables with keys/constraints from lecture slides.

Question Structure Patterns

- Often asks for **SELECT queries** based on a given schema.
- **MCQs** testing definition recall or “what does this SQL do?” (present in quizzes/banks).
- **Normalization problems:** Given a table and FDs, determine normal form or decompose (common in quizzes).
- Joins typically involve writing or identifying correct SQL in E-R context.

Difficulty Distribution

- **Easy:** Definitions, simple SELECT queries (similar to lecture examples).
- **Medium:** JOIN queries, aggregate queries with GROUP BY/HAVING (require understanding from lectures 6–7).
- **Hard:** Complex SQL with multiple joins or subqueries, normalization decomposition (requires multi-step logic).

Repeated Concepts

- SQL clauses (`SELECT`, `WHERE`, `GROUP BY`, `HAVING`).
- ER-to-relational mapping (especially for weak entities or ISA hierarchies).
- Functional dependencies (identified in lectures and likely in exams).

Analysis based on lecture coverage and question bank patterns; actual past exam data not textually available, but these topics align with course emphasis.

Lecture-by-Lecture MCQ Bank

Lecture 1 (Introduction to Databases)

Lecture Topics: Database vs. file systems, DBMS architecture, data models (hierarchical, network, relational), data independence, three-level architecture.

MCQs from Examples/Theory:

1. **Definition-based:** "Which term refers to a self-describing collection of integrated records?"
   A) Database
   B) Schema
   C) Data Model
   D) DBMS
   Correct Answer: A
   Explanation: A *database* is a collection of integrated records stored in a DBMS (Lecture 1).

2. **Conceptual:** "Which of the following is NOT a level of ANSI-SPARC database architecture?"
   A) Internal Level
   B) Conceptual Level
   C) External Level
   D) Transaction Level
   Correct Answer: D
   Explanation: The ANSI-SPARC model has *internal*, *conceptual*, and *external* levels. There is no "transaction level" (Lectures 1–2).

3. **SQL Concept:** "Which SQL command is used to define the structure of a database?"
   A) INSERT
   B) SELECT
   C) CREATE
   D) UPDATE
   Correct Answer: C
   Explanation: `CREATE` is a DDL command used to define database structures (Lecture 5).

4. **Data Independence:** "Changing the logical schema without affecting the external schemas exemplifies:"
   A) Physical Data Independence

B) Logical Data Independence
C) Data Redundancy
D) Data Security
Correct Answer: B
Explanation: *Logical data independence* is the ability to change the conceptual schema without altering external views (Lecture 1).

MCQs from Instructor Patterns: (basics about DBMS functions, disadvantages of file system)

1. **Matching:** "One advantage of DBMS over file systems is..."
A) Increased data redundancy
B) Centralized control of data
C) Larger application size
D) Manual access control
Correct Answer: B
Explanation: A DBMS provides centralized data control and reduces redundancy compared to file systems (Lecture 1).

Lecture 2 (Relational Model, Keys, Constraints)

Lecture Topics: Relational model concepts, tables, schema vs. instance, keys (superkey, candidate, primary), foreign keys, integrity constraints, views.

MCQs from Examples/Theory:

1. **Key Definitions:** "What is a candidate key?"
A) An attribute or set of attributes that uniquely identifies a tuple (not necessarily minimal)
B) The chosen key used to identify tuples
C) A superkey with no redundant attributes
D) A key that references another table
Correct Answer: C
Explanation: A *candidate key* is a minimal superkey (no proper subset is also a superkey) (Lecture 2).

2. **Primary Key:** "Which statement about primary keys is true?"
A) They may contain NULL values.
B) There can be multiple primary keys in one table.
C) They uniquely identify tuples and are chosen from candidate keys.
D) They are always composite keys.
Correct Answer: C

Explanation: A primary key is selected from candidate keys to uniquely identify tuples and cannot contain NULL (Lecture 2).

3. **Foreign Key:** "A foreign key in relation R references..."
   A) The primary key of R itself.
   B) The primary key of another (or the same) relation.
   C) Any unique attribute in any table.
   D) The superkey of R.
   Correct Answer: B
   Explanation: A foreign key in R refers to the primary key of another (or possibly the same) relation to enforce referential integrity (Lecture 2).

4. **Unique vs. Superkey:** "Every candidate key is a superkey, but not every superkey is a candidate key because..."
   A) Superkeys must be composite.
   B) Candidate keys cannot contain NULL, superkeys can.
   C) Candidate keys must be minimal; superkeys may have extra attributes (Lecture 2).
   D) Superkeys only apply to foreign keys.
   Correct Answer: C
   Explanation: Superkeys may have redundant attributes; candidate keys have none (Lecture 2).

5. **Null Values:** "Which of these is a feature of NULLs in a database?"
   A) NULL equals empty string.
   B) NULL values are ignored by aggregate functions like COUNT.
   C) NULL means the attribute must be a foreign key.
   D) NULL values can violate a NOT NULL constraint.
   Correct Answer: B
   Explanation: Aggregates typically ignore NULLs; they are treated as unknown, not equal to anything (Lecture 2 note on NULL).

MCQs from Instructor Patterns: (views, DDL vs DML)

1. **SQL Views:** "Which SQL feature allows a user to see a custom subset of a database?"
   A) Index
   B) View
   C) Trigger
   D) Procedure
   Correct Answer: B

Explanation: A *view* is a virtual table defined by a query, providing a customized perspective on data (Lecture 2).

2. **DDL vs DML:** "Which command is NOT an example of DML?"
   A) SELECT
   B) UPDATE
   C) DELETE
   D) CREATE
   Correct Answer: D
   Explanation: `CREATE` is DDL (defines schema), whereas SELECT/UPDATE/DELETE are DML (manipulate data) (Lecture 5).

Lecture 3 (ER Model Basics)

Lecture Topics: Entity-relationship diagrams, entity sets, attributes, relationship sets, keys, cardinalities (1:1, 1:N, M:N), weak entities, ER-to-relational mapping.

MCQs from Examples/Theory:

1. **ER Components:** "In an ER diagram, a multi-valued attribute is represented by…"
   A) A dashed oval
   B) A double oval (Lecture 3)
   C) A diamond
   D) A double rectangle
   Correct Answer: B
   Explanation: Multi-valued attributes are shown as **double ovals** in ER diagrams (Lecture 3).

2. **Weak Entity:** "A weak entity set cannot be uniquely identified without…"
   A) A primary key
   B) A foreign key from a related strong entity (Lecture 3)
   C) A multi-valued attribute
   D) A derived attribute
   Correct Answer: B
   Explanation: A weak entity depends on a strong (owner) entity for identification and uses the owner's primary key along with its partial key (Lecture 3).

3. **Partial Key:** "What is a partial key (discriminator)?"
   A) A foreign key referencing another table

B) A key that is only partially filled (NULLs allowed)
C) A set of attributes that uniquely identify weak entities related to the same owner (Lecture 3)
D) Any attribute that is not part of the primary key
Correct Answer: C
Explanation: A partial key (or discriminator) uniquely identifies weak entity instances linked to one owner entity (Lecture 3).

4. **Relationship Cardinality:** "A one-to-many (1:N) relationship from A to B means…"
   A) One B is related to many A's
   B) One A is related to many B's (Lecture 3)
   C) Each A is related to at most one B
   D) Many A's relate to many B's
   Correct Answer: B
   Explanation: In 1:N from A to B, one A can be associated with many B's; each B associates with at most one A (Lecture 3).

5. **ER-to-Relational:** "In mapping a many-to-many (M:N) relationship to tables, we…"
   A) Add a foreign key to one of the entity tables
   B) Create a new relation (table) for the relationship including foreign keys to both entities (Lecture 3)
   C) Merge the two entity tables
   D) Cannot represent M:N in relational model
   Correct Answer: B
   Explanation: An M:N relationship is implemented by creating a new table with foreign keys referencing the primary keys of both participating entities (common DB design rule).

MCQs from Instructor Patterns: (ISA and specialization)

1. **Specialization:** "Specialization in ER modeling refers to…"
   A) Combining several entities into one union type (Lecture 4)
   B) Defining a new entity as a subtype of an existing entity (Lecture 4)
   C) Merging attributes across entities
   D) Creating a multi-valued attribute
   Correct Answer: B
   Explanation: Specialization (or subclassing) defines sub-entities (subclasses) under a parent entity, inheriting attributes (Lecture 4).

2. **Generalization:** "Generalization is best described as…"
   A) Partitioning an entity into subgroups
   B) Creating a higher-level entity from shared attributes of lower-level entities (Lecture 4)
   C) The process of normalization
   D) Assigning data types to attributes
   Correct Answer: B
   Explanation: Generalization abstracts shared attributes of several entities into a superclass (Lecture 4).

Lecture 4 (EER Extensions)

Lecture Topics: Specialization/generalization, union types (categories), constraints on hierarchies, ER diagram enhancements (Lecture 4).

MCQs from Examples/Theory:

1. **Category (Union Type):** "A union type (category) in EER modeling…"
   A) Is a subtype of a given entity (generalization)
   B) Is an entity that contains all attributes of two or more entities not in a superclass (Lecture 4)
   C) Is another name for aggregation
   D) Is the same as specialization
   Correct Answer: B
   Explanation: A category (union type) is a subclass that includes entities from multiple distinct superclasses (Lecture 4).

2. **ISA Constraints:** "In an ISA hierarchy with total participation, every instance of the superclass…"
   A) Must belong to at least one subclass (Lecture 4)
   B) May belong to zero subclasses
   C) Cannot have its own primary key
   D) Must be a weak entity
   Correct Answer: A
   Explanation: Total specialization means each superclass instance is a member of some subclass (Lecture 4).

3. **Multi-Level ISA:** "Which is a correct description of multi-level generalization?"
   A) Generalizing an entity over three or more levels (Lecture 4)
   B) Specializing a subclass further into its own subclasses

C) Combining specialization and union
D) Not possible in EER
Correct Answer: B
Explanation: Multi-level inheritance means a subclass can further specialize into its own subclasses (Lecture 4 shows multilayer examples).

4. **Aggregation (if covered):** "Aggregation in ER modeling is used to…"
   A) Combine two weak entities
   B) Model a relationship as an entity in higher-level relationships (Lecture 4 concept)
   C) Normalize tables
   D) Expand composite attributes
   Correct Answer: B
   Explanation: Aggregation treats a relationship set as a higher-level entity, often used when an n-ary relationship participates in another relationship (Lecture 4 context).

MCQs from Instructor Patterns: (category vs specialization)

1. **Category vs. Specialization:** "The difference between a union type and specialization is that a union type…"
   A) Is a subset of a single entity
   B) Inherits from multiple parent entities (Lecture 4)
   C) Cannot have relationships of its own
   D) Has no key
   Correct Answer: B
   Explanation: A union type (category) has multiple parent entity sets, whereas specialization has one parent (Lecture 4).

Lecture 5 (SQL Basics – SELECT, DDL/DML)

Lecture Topics: SQL overview, DDL vs DML, SELECT-FROM-WHERE structure, basic queries, NULL, DISTINCT, logical operators.

MCQs from Examples/Theory:

1. **SQL Structure:** "Which clause is mandatory in an SQL SELECT statement?"
   A) WHERE
   B) HAVING
   C) FROM (Lecture 5)
   D) GROUP BY
   Correct Answer: C

Explanation: `FROM` (to specify tables) and `SELECT` (columns) are mandatory. `WHERE`, `GROUP BY`, etc. are optional (Lecture 5).

2. **Logical Operators:** "In SQL, the AND operator..."
   A) Has lower precedence than OR
   B) Combines two conditions and returns true only if both are true (Lecture 5)
   C) Is only used in UPDATE statements
   D) Is evaluated after BETWEEN and IN
   Correct Answer: B
   Explanation: `AND` requires both conditions to be true. (Lecture 5 covers logical AND, OR, NOT).

3. **IN vs. ANY:** "`column IN (value1, value2)` is functionally equivalent to..."
   A) `column = value1 OR column = value2` (Lecture 5 concept)
   B) `column > ALL (subquery)`
   C) `column LIKE 'value1'`
   D) `column BETWEEN value1 AND value2`
   Correct Answer: A
   Explanation: `IN` checks if the column matches any listed value (Lecture 5 notes on logical operators).

4. **NULL Behavior:** "Which SQL comparison yields TRUE if `Salary` is NULL?"
   A) `Salary = NULL`
   B) `Salary <> NULL`
   C) `Salary IS NULL` (Lecture 5)
   D) `Salary != 0`
   Correct Answer: C
   Explanation: Only `IS NULL` checks for NULL; = or <> with NULL yields UNKNOWN (Lecture 5 on NULLs).

5. **DISTINCT:** "What does `SELECT DISTINCT Dept FROM Employee;` return?"
   A) All department values including duplicates
   B) Only employees from the first department
   C) Each distinct department value once (Lecture 8)
   D) An error, DISTINCT cannot be used without WHERE
   Correct Answer: C
   Explanation: `DISTINCT` removes duplicates, so each department appears once (Lecture 8 example).

MCQs from Examples: (based on actual query in slides)

1. **Based on Lecture Example:** “Given tables EMPLOYEE(SSN, Fname, Lname, Dno) and DEPARTMENT(Dnumber, Dname), what does the query `SELECT Fname, Lname FROM EMPLOYEE WHERE Dno = 5;` do?”
   A) Lists all employees from department 5 (Lecture 5 pattern)
   B) Updates department number to 5
   C) Deletes employees in department 5
   D) Lists all departments that have an employee
   Correct Answer: A
   Explanation: `WHERE Dno = 5` filters employees with Dno=5; SELECT returns their first and last names (SQL basics).

2. **SQL Command Category:** “Which of the following is a DML command?”
   A) ALTER
   B) DROP
   C) INSERT (Lecture 5)
   D) CREATE
   Correct Answer: C
   Explanation: `INSERT` is DML (manipulates data), while `ALTER`, `DROP`, `CREATE` are DDL (schema changes) (Lecture 5).

Lecture 6 (SQL – Aggregation, Group By, Functions)

Lecture Topics: Aggregate functions (`SUM`, `AVG`, `COUNT`), `GROUP BY`, `HAVING`, subqueries basics, updating multiple tables.

MCQs from Examples/Theory:

1. **Aggregate Use:** “What is the output of `SELECT COUNT(*), SUM(Salary) FROM Employee;`?”
   A) The number of employees and the total salary sum (Lecture 6)
   B) The average salary of employees
   C) Employee with highest salary
   D) All individual salaries
   Correct Answer: A
   Explanation: `COUNT(*)` returns number of rows; `SUM(Salary)` sums all salaries (Lecture 6 example).

2. **GROUP BY:** “Which query correctly shows average salary by department?”
   A) `SELECT Dno, AVG(Salary) FROM Employee;`
   B) `SELECT AVG(Salary) FROM Employee GROUP BY Dno;` (Lecture 6)
   C) `SELECT Dno, SUM(Salary) FROM Employee GROUP BY AVG;`

D) `SELECT Dno FROM Employee GROUP BY Salary;`
Correct Answer: B
Explanation: Must include `GROUP BY Dno` to aggregate by department, selecting AVG(Salary) (Lecture 6).

3. **HAVING Clause:** "HAVING is used to…"
   A) Filter rows before aggregation
   B) Filter groups after aggregation (Lecture 6)
   C) Order the result set
   D) Remove duplicates
   Correct Answer: B
   Explanation: `HAVING` applies conditions on aggregated groups, unlike `WHERE` which filters individual rows (Lecture 6).

4. **NULL in Aggregates:** "If a column contains NULLs, `AVG()` will…"
   A) Return NULL
   B) Treat NULLs as zero
   C) Ignore NULL values in the average (Lecture 6)
   D) Cause an error
   Correct Answer: C
   Explanation: Aggregate functions (like AVG) ignore NULLs in their calculations (SQL behavior).

5. **Update Multiple Rows:** "What does `UPDATE Employee SET Salary = Salary * 1.1 WHERE Dept = 'Sales';` do?"
   A) Gives all employees a 10% raise
   B) Gives employees in Sales department a 10% raise (Lecture 6 UPDATE example)
   C) Deletes sales department employees
   D) Changes department to 'Sales' where salary increased
   Correct Answer: B
   Explanation: The `WHERE Dept = 'Sales'` restricts the 10% increase to only Sales employees (Lecture 6 UPDATE usage).

MCQs from Examples:

1. **Subquery (if covered):** "Given DEPT(DeptID, Location) and EMP(EmpID, DeptID, Salary), which query lists employees with salary above their department's average?"
   A) `SELECT EmpID FROM EMP WHERE Salary > (SELECT AVG(Salary) FROM EMP);`
   B) `SELECT EmpID FROM EMP WHERE Salary > (SELECT AVG(Salary) FROM`

`EMP WHERE DeptID = EMP.DeptID);` (Lecture 8 style)
C) `SELECT EmpID FROM EMP GROUP BY DeptID HAVING Salary > AVG(Salary);`
D) `SELECT EmpID FROM EMP, DEPT WHERE EMP.Salary > AVG(DEPT.Salary);`
Correct Answer: B
Explanation: Subquery correlates on DeptID to compare each employee's salary to the average of their own department (concept from Lecture 8 subqueries).

Lecture 7 (SQL – Joins, Set Operations)

Lecture Topics: JOIN types (`INNER`, `OUTER`, `CROSS`, `NATURAL`), UNION, INTERSECT, set difference, complex queries.

MCQs from Examples/Theory:

1. **INNER JOIN:** "What is true about an INNER JOIN between A and B?"
   A) It returns all rows from A, matching or not.
   B) It returns rows only where A and B have matching values (Lecture 7).
   C) It returns all rows from A and B regardless of matches.
   D) It duplicates rows from A for each in B.
   Correct Answer: B
   Explanation: `INNER JOIN` returns only matching rows from both tables (Lecture 7).

2. **LEFT JOIN:** "A LEFT OUTER JOIN will include..."
   A) Only rows where both tables match
   B) All rows from left table and matching right rows (Lecture 7)
   C) All rows from right table and matching left rows
   D) No rows if there is no match
   Correct Answer: B
   Explanation: A LEFT OUTER JOIN includes every row from the left table, with NULLs for non-matches in the right table (Lecture 7).

3. **NATURAL JOIN:** "A NATURAL JOIN automatically uses..."
   A) All columns in both tables for matching.
   B) Matching columns with the same names in both tables (Lecture 7).
   C) The first column of each table.
   D) Only the primary keys.
   Correct Answer: B

Explanation: A natural join matches on all columns that have the same name in both tables (Lecture 7).

4. **UNION:** "Which statement about UNION is correct?"
   A) It returns duplicate rows (no)
   B) It combines query results, eliminating duplicates (Lecture 7)
   C) It requires both queries to have different number of columns
   D) It sorts the results in descending order automatically
   Correct Answer: B
   Explanation: `UNION` merges two result sets and removes duplicates (Lecture 7).

5. **INTERSECT vs MINUS:** "What does `INTERSECT` do?"
   A) Returns rows in the first query but not the second
   B) Returns rows common to both queries (Lecture 7)
   C) Returns rows in either query (like UNION)
   D) Returns all rows from the second query only
   Correct Answer: B
   Explanation: `INTERSECT` returns only rows that appear in both result sets.

MCQs from Examples:

1. **Query Understanding:** "Given tables Orders(OrderID, CustID) and Customers(CustID, Name), which SQL returns all customers who have made at least one order?"
   A) `SELECT Name FROM Customers` (missing condition)
   B) `SELECT Name FROM Customers WHERE CustID IN (SELECT CustID FROM Orders);` (Lecture 7 style)
   C) `SELECT DISTINCT Customers.Name FROM Customers, Orders;` (no join condition)
   D) `SELECT Name FROM Customers LEFT JOIN Orders;`
   Correct Answer: B
   Explanation: Uses a subquery on Orders to filter customers; correct pattern (Lecture 7 had similar subquery usage).

2. **SET Theory:** "What does `SELECT A.ID FROM A UNION SELECT B.ID FROM B;` produce?"
   A) All IDs from A and B, including duplicates
   B) All unique IDs appearing in either A or B (Lecture 7)
   C) Only IDs that appear in both A and B
   D) IDs from A only
   Correct Answer: B

Explanation: `UNION` combines IDs from both tables, removing duplicates (Lecture 7).

Lecture 8 (SQL – Subqueries, ANY/ALL)

Lecture Topics: Nested subqueries, correlated subqueries, ANY/ALL/EXISTS, query evaluation.

MCQs from Examples/Theory:

1. **Subquery ANY:** "In `WHERE Salary > ANY (SELECT Salary FROM Dept WHERE DeptName='Sales')`, ANY means…"
   A) Salary must be greater than all values in subquery
   B) Salary must be greater than at least one value returned (Lecture 8)
   C) Salary must equal some value in subquery
   D) Salary must be greater than any possible number
   Correct Answer: B
   Explanation: `> ANY` is true if the Salary is greater than at least one value from the subquery (Lecture 8).

2. **Subquery ALL:** "In contrast, `Salary > ALL (…)` means…"
   A) Salary is greater than every value returned by subquery (Lecture 8)
   B) Salary equals all values (impossible)
   C) Salary less than any value
   D) Salary in the subquery results
   Correct Answer: A
   Explanation: `> ALL` requires Salary to be larger than all values produced by the subquery (Lecture 8).

3. **Correlated Subquery:** "Which is a correlated subquery example?"
   A) `SELECT * FROM Employee WHERE Salary > (SELECT AVG(Salary) FROM Employee);`
   B) `SELECT * FROM Employee e WHERE Salary > (SELECT AVG(Salary) FROM Employee WHERE Dept = e.Dept);` (Lecture 8)
   C) `SELECT * FROM Employee WHERE Dept = ANY(1,2,3);`
   D) `SELECT * FROM Employee, Department;`
   Correct Answer: B
   Explanation: The inner query references the outer query alias `e.Dept`, making it correlated (Lecture 8 concept).

4. **EXISTS:** "`WHERE EXISTS (SELECT * FROM Dept WHERE DeptID = Employee.DeptID)` returns true when…"
   A) The subquery returns any row (Lecture 8)
   B) The subquery is empty
   C) The department has no employees
   D) All departments match
   Correct Answer: A
   Explanation: `EXISTS` is true if the subquery finds at least one matching row (SQL fundamentals).

5. **ANY with comparison:** "`Salary >= ANY (SELECT Bonus FROM Employee)` is equivalent to…"
   A) `Salary >=` the lowest bonus value (Lecture 8)
   B) `Salary >=` the highest bonus value
   C) `Salary >=` all bonus values
   D) An error, misuse of ANY
   Correct Answer: A
   Explanation: `>= ANY` means Salary is greater than or equal to at least one returned Bonus; effectively >= the minimum (Lecture 8 logic).

MCQs from Examples:

1. **Query Behavior:** "What does `SELECT Name FROM Employee WHERE DeptID = ANY (SELECT DeptID FROM Department WHERE Location='London')` do?"
   A) Lists names of all employees in London-based departments (Lecture 8 structure)
   B) Lists names of London employees only
   C) Lists all employees with any department ID
   D) Lists employees not in London
   Correct Answer: A
   Explanation: Subquery finds DeptIDs in London; ANY matches if Emp's DeptID is any of those (nested query logic).

Lecture 9 (Normalization Part I)

Lecture Topics: Functional dependencies, anomalies, definitions of 1NF–3NF, BCNF, examples of violations.

MCQs from Examples/Theory:

1. **1NF Rule:** “A table is in First Normal Form (1NF) if…”
   A) It has no composite keys.
   B) All attributes have atomic values (Lecture 10).
   C) It has no foreign keys.
   D) It contains no transitive dependencies.
   Correct Answer: B
   Explanation: 1NF requires that each column holds atomic (indivisible) values (Lecture 10).

2. **2NF Rule:** “A table in 1NF is in 2NF if…”
   A) Every non-key attribute depends on the whole primary key, not part of it (Lecture 10).
   B) It has no duplicate rows.
   C) It has no NULL values.
   D) It has a composite key.
   Correct Answer: A
   Explanation: 2NF eliminates partial dependencies; every non-key attribute must fully depend on the entire key (Lecture 10).

3. **3NF Rule:** “To achieve Third Normal Form (3NF), a relation must…”
   A) Be in 2NF and have no transitive dependencies (Lecture 10).
   B) Remove all foreign keys.
   C) Have only one candidate key.
   D) Contain no derived attributes.
   Correct Answer: A
   Explanation: 3NF requires no non-key attribute is transitively dependent on the primary key (Lecture 10).

4. **Transitive Dependency:** “Which is an example of a transitive dependency?”
   A) A → B and B → C implies A → C (Lecture 10 concept).
   B) A → B where A is composite.
   C) A foreign key dependency.
   D) A multi-valued attribute in a table.
   Correct Answer: A
   Explanation: Transitive FD is when A determines B and B determines C, hence A determines C indirectly (Lecture 10).

5. **BCNF:** “Boyce-Codd Normal Form differs from 3NF because…”
   A) It is weaker than 3NF.
   B) It requires every determinant to be a key (Lecture 9 mention).
   C) It allows partial dependencies.

D) It only applies to NoSQL databases.
Correct Answer: B
Explanation: BCNF is stricter; every non-trivial FD X→Y implies X is a superkey (Lecture 9 list of normal forms).

MCQs from Examples:

1. **Normalization Example:** "Given table R(A, B, C) with functional dependencies A→B and B→C, which normal form does R violate?"
   A) 1NF
   B) 2NF
   C) 3NF (Lecture 10)
   D) BCNF
   Correct Answer: C
   Explanation: A→B and B→C creates a transitive dependency A→C; R is in 2NF but not 3NF (3NF violation).

2. **Dependency Identification:** "In relation R(X, Y, Z) with primary key (X, Y), a dependency X→Z is..."
   A) Full functional dependency (no, depends on subset of key)
   B) Partial dependency (Lecture 10)
   C) Transitive dependency
   D) Not a dependency
   Correct Answer: B
   Explanation: Z depends on part of the composite key X; this is a partial dependency (Lecture 10).

Lecture 10 (Normalization Part II)

Lecture Topics: Detailed rules of 1NF, 2NF, 3NF, decomposition, examples of normalizing tables.

MCQs from Examples/Theory:

1. **Atomic Rule:** "Rule 1 of 1NF states that..."
   A) Each column must be of integer type.
   B) Each column must have atomic (indivisible) values (Lecture 10).
   C) There can be no more than 10 columns.
   D) Each row must have a unique identifier.
   Correct Answer: B

Explanation: 1NF Rule 1 requires atomic values (no lists or sets in a column) (Lecture 10).

2. **Repeating Groups:** "Rule 2 of 1NF means a table…"
   A) Must have no composite attributes.
   B) Must not contain repeating groups of columns (Lecture 10).
   C) Must have no NULL values.
   D) Must be in 2NF automatically.
   Correct Answer: B
   Explanation: No repeating group columns ensures there is no redundancy of columns (Lecture 10).

3. **Composite Key and 2NF:** "A table with a single-attribute key is…"
   A) Never in 2NF.
   B) Automatically in at least 2NF (Lecture 10 note).
   C) Automatically in BCNF.
   D) Automatically in 1NF only.
   Correct Answer: B
   Explanation: If the primary key is atomic (single-column), there can be no partial dependency, so it's in 2NF (Lecture 10 note).

4. **Transitive Dependency:** "If A→B and B→C, then C is transitively dependent on A if…"
   A) A is not a key.
   B) B is not a key (Lecture 10 understanding).
   C) C is prime attribute.
   D) A, B, C are composite keys.
   Correct Answer: B
   Explanation: If A determines B and B determines C and neither A→C nor B is key, then C is transitively dependent on A (Lecture 10 concept).

5. **Decomposition:** "To normalize a relation, we decompose it so that each table…"
   A) Contains only one attribute.
   B) Has an index on every column.
   C) Satisfies the desired normal form without losing dependencies (implied Lecture 10 process).
   D) Has no keys.
   Correct Answer: C
   Explanation: Decomposition should preserve dependencies and eliminate anomalies, resulting in normalized tables (normalization procedure).

MCQs from Examples:

1. **Normalization Scenario:** "Table R(A,B,C,D) has FDs: A,B→C and C→D. Which of the following decompositions achieves 3NF?"
   A) R1(A,B,C), R2(A,B,D)
   B) R1(A,B,C), R2(C,D) (Lecture 10 pattern)
   C) R1(A,D), R2(B,C)
   D) R1(A,B), R2(C,D)
   Correct Answer: B
   Explanation: R1 with (A,B,C) covers A,B→C, and R2(C,D) covers C→D; both are in 3NF and preserve dependencies (standard decomposition).

2. **Violation Check:** "Given R(X,Y,Z) with key (X,Y) and FD X→Z, R is currently in 1NF. The FD X→Z implies…"
   A) R is already in 2NF
   B) R violates 2NF (Lecture 10)
   C) R violates BCNF
   D) R has no violation
   Correct Answer: B
   Explanation: Z depends only on part (X) of the composite key, a partial dependency (2NF violation).

Lecture 11 (Query Processing & Optimization)

Lecture Topics: Query processing steps (scanning, parsing, optimization), relational algebra (σ, π, ×), query tree evaluation, cost considerations.

MCQs from Examples/Theory:

1. **Relational Algebra σ (selection):** "σ_{Department='Sales'} (Employee)" selects…"
   A) Columns from Employee table.
   B) Tuples of Employee where Department = 'Sales' (Lecture 11)
   C) All unique departments.
   D) Employees working in any department.
   Correct Answer: B
   Explanation: σ (selection) filters rows; here Department='Sales' yields only those employees (Lecture 11 relational algebra).

2. **Relational Algebra π (projection):** "π_{Name,Salary} (Employee)" means…"

A) Select rows with name and salary only.
B) Returns a table with only the Name and Salary columns from Employee (Lecture 11).
C) Returns all employees named Salary.
D) Combines Name and Salary into one column.
Correct Answer: B
Explanation: π (projection) selects specified columns (attributes) from a relation (Lecture 11).

3. **Cartesian Product:** "Relational algebra's × (cross join) is 'meaningless' without..."
A) A subsequent selection or condition (Lecture 11 example).
B) At least three tables.
C) Aggregate function.
D) Indexes.
Correct Answer: A
Explanation: Cross product followed by selection simulates a join; alone it returns every pair of rows (Lecture 11 notes).

4. **Query Processing:** "In query processing, the parser's role is to..."
A) Optimize execution plan.
B) Check query syntax (Lecture 11).
C) Execute the query.
D) Fetch data from disk.
Correct Answer: B
Explanation: The parser checks SQL syntax and generates an internal representation (Lecture 11).

5. **Query Optimization:** "One goal of query optimization is to minimize..."
A) The number of tables.
B) The number of joins in SQL text.
C) The cost of executing the query (e.g., I/O, CPU) (Lecture 11 concept).
D) The size of the result set.
Correct Answer: C
Explanation: Optimization finds efficient execution plans to reduce resource usage (Lecture 11 discussion).

MCQs from Examples:

1. **Algebra Example:** "Given Employee(E) and Department(D) relations, what does `π_{E.Name, D.Location}(E ⋈_{E.Dno=D.Dnumber} D)` produce?"
A) All names of employees (and all locations)